Answer Key

Grade 6 Mathematics · Unit 8: Equations and Inequalities · Answers for every item on Sheets 1 and 2 of each lesson packet. Constructed responses show a model answer; accept equivalent reasoning.

LESSON 8-1

Equations and Solutions · Writing Equations

Sheet 1 · Equations and Solutions

1. a. T (20 = 20) b. F (30 ≠ 35) c. T (16 = 16) d. F (8 ≠ 7) e. T (30 = 30) f. T (54 = 54) g. T (4 = 4) h. F (72 ≠ 63)
2. x = 25. Checks: 15 + 16 = 31 (no); 25 + 16 = 41 (yes); 35 + 16 = 51 (no).
3. b = 11: 11 − 7 = 4, and 4 ≠ 18, so no. b = 25: 25 − 7 = 18, so yes. The solution is b = 25.
4. 4p means 4 × p, not 4 + p. Substituting gives 4 × 8 = 32, and 32 ≠ 12, so p = 8 is not a solution. The solution is p = 3, because 4 × 3 = 12.
5. t = 36: 6 × 36 = 216, and 216 ≠ 42, so no. t = 7: 6 × 7 = 42, so yes. The solution is t = 7 tickets.

Sheet 2 · Writing Equations

1. a. n + 9 = 31 b. 7n = 63 c. n − 12 = 20 d. n ÷ 5 = 8
2. s − 14 = 29. Test: 43 − 14 = 29, so s = 43 is the solution.
3. 4v = 56. Test 12: 4 × 12 = 48 (no). Test 14: 4 × 14 = 56 (yes). v = 14 students.
4. 5p = 85. Test: 5 × 17 = 85, so p = 17 pages a day is the solution.
5. Omar's equation gives b = 17, which would make the brother older than Tia. The brother's age plus 4 is Tia's age: b + 4 = 13, so b = 9.

TEST PRACTICE

6. B – 9w = 72
7. C – n = 42

LESSON 8-2

Addition Equations · Subtraction Equations

Sheet 1 · Addition Equations

1. a. x = 8 b. n = 18 c. y = 35 d. m = 37 e. k = 5.5 f. w = 4.25
2. p + 46 = 93; p = 47
3. b + 27 = 64; b = 37 books. Check: 37 + 27 = 64.
4. 38.50 + m = 52; m = $13.50. Check: 38.50 + 13.50 = 52.
5. No. n = 32 is correct. Dante checked with the wrong operation. He should substitute 32 into the original equation: 32 + 18 = 50, which is true.

Sheet 2 · Subtraction Equations

1. a. n = 45 b. k = 85 c. p = 9.5
2. t − 54 = 35 (or t − 35 = 54); t = 89
3. g − 18.25 = 31.75; g = $50.00
4. c − 26 = 59; c = 85 cars. Check: 85 − 26 = 59.
5. Leo. 84 − 24 = 60 is true. Mia subtracted instead of adding: 36 − 24 = 12, not 60.

TEST PRACTICE

6. C – p = 74
7. B – m − 48 = 37

LESSON 8-3

Multiplication Equations · Division Equations

Sheet 1 · Multiplication Equations

1. a. x = 9 b. n = 8 c. y = 12 d. k = 7 e. m = 17 f. w = 1.5
2. 5c = 65; c = 13
3. 5f = 11.25; f = $2.25. Check: 5 × 2.25 = 11.25.
4. 6s = 84; s = 14 seedlings. Check: 6 × 14 = 84.
5. 4n = 80 has the greater solution. The same total split into fewer equal groups makes each group larger. Check: 4n = 80 gives n = 20; 5n = 80 gives n = 16.

Sheet 2 · Division Equations

1. a. n = 42 b. y = 65 c. k = 45
2. t ÷ 3 = 17; t = 51
3. r ÷ 8 = 2.5; r = 20 meters
4. z ÷ 6 = 2; z = 12 slices. Check: 12 ÷ 6 = 2.
5. The cards are shared by 4 people (Kayla and 3 friends), not 3. Correct equation: c ÷ 4 = 9, so c = 36 cards.

TEST PRACTICE

6. D – y = 54
7. A – p ÷ 12 = 5

LESSON 8-4

Writing Inequalities · Representing Inequalities

Sheet 1 · Writing Inequalities

1. a. x > 12 b. x ≤ 30 c. x < 8 d. x ≥ 15 e. x ≤ 6 f. x > −3 g. x < 0 h. x ≤ 100
2. a. h ≥ 54 b. s ≤ 35 c. n > 10 d. p < 120
3. r ≤ 48. Yes, 48 riders is allowed because 48 ≤ 48 is true. No, 50 riders is not allowed because 50 ≤ 48 is false.
4. Sample: A movie must be shorter than 60 minutes (t < 60). Solutions: 45 and 59.5 (any number less than 60).
5. No. "At most 10" is x ≤ 10, so 10 is a solution. "Less than 10" is x < 10, so 10 is not a solution.

Sheet 2 · Representing Inequalities

1. x > −1
2. Closed dot at 2, arrow right. Sample solutions: 2 and 5 (any number 2 or greater).
3. a. x > 3 b. x ≤ −2
4. w ≤ 24. Closed dot at 24, arrow left.
5. The dot should be open. > does not include the boundary: 5 > 5 is false, so 5 is not a solution. (His arrow to the right is correct.)

TEST PRACTICE

6. B – x ≤ 4
7. D – 0

Sheet 1 · Graphing Inequalities

1. a. closed, right b. open, left c. closed, left d. open, right e. closed, right f. open, left g. open, right h. closed, left

2. Closed dot at −1, arrow left. Sample solution: −3 (any number −1 or less).

3. $a \geq 10$. Closed dot at 10, arrow right.

4. $m < 6$. Sample: A small boat can hold fewer than 6 people.

Sheet 2 · Testing Solutions to Inequalities

1. $x > 7$: no, no, yes. $x \leq 7$: yes, yes, no. $x \geq 4$: yes, yes, yes.

2. a. no ($9 < 9$ is false) b. yes ($0 \geq -1$) c. no ($3 \leq 2.5$ is false)

3. $s > 450$. $450 > 450$ is false; $445 > 450$ is false; $472 > 450$ is true. Only 472 wins.

4. $-3 \leq -1$ is true; $0 \leq -1$ is false; $2 \leq -1$ is false. Only −3 is a solution. Graph: closed dot at −1, arrow left.

5. Every number less than 12 is a solution, including decimals like 11.5 and negative numbers like −3. There are infinitely many solutions.

TEST PRACTICE

6. A – −3

7. D – 15.5